Calculation

2025 GHGtarget = 91.16*(1-2%)=89,337 gCO2eq/MJ

2030 GHGtarget = 91.16*(1-6%)=85,69 gCO2eq/MJ

Handymax

DKAAR -> GBABD -> NLRTM -> DKAAR

DWT : 40 799 T

Distance : 1521 n miles

Travel time : 144.9h

Emission : 352 263.6 kg CO2eq2026

CII score = 5.68

EEXI = CO2 / (DWT*speed) = 352263.6*10^3 / (48 000*14) =524.202 g CO2/(t.kn)

EU ETS are calculated on a basic price, in 2026 it was 80€ (it's an average in the 2026 year) per tonne of CO2, before we said that we assumed that in 2026 the emissions are 100% covered.

2024 EU ETS : 65*(352,263* 0,4)=9 158,838 €

2025EU ETS : 73,43*(352,263*0,7) =18 106,67 €

2026 EU ETS : 80*352,263 =28 181,04 €

Energy used (MJ) = Emission(g) / GHG actual

Energy used =352 263*10^3/91,744=3 839 629,84 MJ

Compliance Balance (gCO2eq) = (GHG target - GHG actual) × Energy used

Compliance = (89,337-91,744)*3839629,84=-9 241 989,0249 gC02 DEFICIT

Penalty (€) = |Compliance Balance| / GHG target × (2400 / 41 000)

Penalty = 9241989,0249/89,337 *(2400/41000)=6 055,66 €

Panamax

SEGVX -> BEANR -> FIKTK -> SEGVX

DWT : 52 713 T

Distance: 2880 nm

Travel time : 306.4h

Emission : 749 203.2 kg CO2eq

CII = 4.935

EEXI = 749 203.2*10^3 / (75 000*14.5) =688,922 gCO2/(t.kn)

2024 EU ETS : 65*(749,3032 * 0,4)=19 481,883 €

2025EU ETS : 73,43*(749,3032 *0,7) =38 514,934 €

2026 EU ETS : 80*749,3032 =59 944,256 €

Energy used =749303*10^3/91,744=8 167 324,294 MJ

Compliance = (89,337-91,744)*8 167 324,294=-19 658 749,576 gCO2eq DEFICIT

Penalty = 19 658 749,576/89,337 *(2400/41000)=12 881,0691 €

KAMSARMAX

NLRTM ->NOKRS->SEGOT->NLRTM

DWT : 63 073 T

Distance : 1050 nm

Travel time : 94.6h

Emission : 321 153 kg CO2eq

CII : 4.85

EEXI = CO2 / (DWT*speed) = 321 153 10^3 / (82000*14)= 279,75 gCO2 / (t.kn)

2024 EU ETS : 65*(321,153 * 0,4)=8 349,978 €

2025EU ETS : 73,43*(321,153*0,7) =16 507,585€

2026 EU ETS : 80*321,153=25 692,24 €

Energy used =321153*10^3/91,744=3 500 534,0949 MJ

Compliance = (89,337-91,744)*3 500 534,0949=-8 425 785,566 gCO2eq DEFICIT

Penalty = 8 425 785,566/89,337 *(2400/41000)=5 520,856 €

CAPESIZE

BEANR -> SEGVX -> SEGOT -> BEANR

DWT : 220 766 T

Distance : 2489.0 nm

Travel time : 157.5h

Emission : 2 697 727.5 kg CO2eq

CII : 4,91

EEXI = 2 697 727.5*10^3/(180 000*15)= 999.53 g CO2/(t.kn)

2024 EU ETS : 65*(2 697,7275* 0,4)=70 140,915 €

2025EU ETS : 73,43*(2 697,7275*0,7) =138 665,891 €

2026 EU ETS : 80*2 697,7275=215 818,2 €

Energy used =2697727*10^3/91,744=29 404 942,0126 MJ

Compliance = (89,337-91,744)*29 404 942,0126=-70 777 695,424 gCO2eq DEFICIT

Penalty = 70777695,424 /89,337 *(2400/41000)= 46 375,90931€

VLOC

BEANR -> FIKTK -> SEGOT -> BEANR

DWT : 402 347 T

Distance : 2647 nm

Travel time : 171.9 h

Emission : 2 157 305 kg CO2eq

CII : 2.03

EEXI = CO2 / (DWT*speed) = 2157305*10^3/(300000*15)=479,401 gCO2/(t.kn)

2024 EU ETS : 65*(2 157,305* 0,4)=56 089,93 €

2025EU ETS : 73,43*(2 157,305*0,7) =110 887,634 €

2026 EU ETS : 80*2 157,305=172 584,4 €

Energy used =2157305*10^3/91,744=23 514 398,762 MJ

Compliance = (89,337-91,744)*23 514 398,762=-56 599 157,82 gCO2eq DEFICIT

Penalty = 56 599 157,82 /89,337 *(2400/41000)=37 085,65805 €